Vitthal Humbe

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CAREER OBJECTIVE

Third year AI/ML student who has shipped end-to-end software systems on AWS & published to PyPI, with hands-on experience in backend architecture and cloud-native ML pipelines. Led a 4-person team to win Best Pitch Award at COEP's hackathon. Looking to apply this in a Software Engineer role.

EDUCATION

B.Tech, Computer Science & Engineering (Artificial Intelligences And Machine Learning) — 2025 - 2028
Pimpri Chinchwad College of Engineering

Diploma, Artificial Intelligence And Machine Learning — 2022 - 2025
New Polytechnic Kolhapur

Percentage: 95.44%

TRAININGS / CERTIFICATIONS

Introduction To Machine Learning
Jul 2024 - Sep 2024

SWAYAM NPTEL, Virtual

Completed course with silver certification with 77%

Cloud Computing
May 2024 - Jul 2024

BITS TECHNO, Kolhapur

Hands-on training covering AWS EC2, S3, and IAM for cloud deployment and access management Deployed and managed cloud-based applications focusing on scalability and security.

PORTFOLIO

GitHub link ↗

Leetcode link ↗

PROJECTS

ClaimCraft GenAI-powered X12 EDI Parser & Validation System ↗
Apr 2026

- Built a from-scratch Python engine to parse, validate, and audit healthcare X12 EDI files (837P, 837I, 835, 834) with a Pushdown Automaton model for structural compliance.
- Integrated Llama 3.3 via Groq API to generate plain-English error explanations, suggested fixes, and contextual insights for complex HIPAA validation failures.
- Developed an end-to-end platform (FastAPI + React) enforcing 400+ HIPAA 5010 rules with live CMS/NLM API validation, batch processing, and claim reconciliation.

Bitlearn - Custom ML Library with Python API & C++ Backend ↗
Jan 2026 - Feb 2026

- Built a machine learning library from scratch with a Python interface and pybind11-powered C++ backend for performance-critical computations.
- Implemented core algorithms including Linear Regression (GD, SGD, Regularization) and Neural Networks with backpropagation and custom activation functions.
- Published to PyPI with GitHub Actions CI/CD pipeline and cibuildwheel cross-platform wheel builds, tested on real datasets including MNIST.

Hybrid Deep Learning & ML Model for Multi-label Chest X-ray Classification
Sep 2025 - Oct 2025

- Designed a hybrid model combining DenseNet-121 for deep feature extraction and classical ML classifiers (Random Forest, SVM) for multi-label chest X-ray classification.
- Extracted 1024-dimensional feature vectors from DenseNet-121 and trained ensemble classifiers to improve diagnostic accuracy and interpretability.
- Achieved clinically reliable performance (NPV > 0.90), validating suitability for real-world medical screening workflows.

Bery - Own Compiled Programming Language ↗
May 2026 - Present

- Bery is a compiled, statically typed, object-oriented programming language that we are building from scratch as a team of five.
- Uses LLVM as its backend and includes built-in support for core data structures such as stacks, queues, and hashmaps.
- Building the complete ecosystem, including the compiler, toolchain, CLI, documentation website, and a web-based playground and assembler; currently developing toward the v1.0.0 release.

SKILLS

- Python
- Java
- FastAPI
- Next.js
- Node.js
- React
- TensorFlow
- scikit-learn
- Linux
- MongoDB
- MySQL
- Amazon Web Services (AWS)

ADDITIONAL DETAILS

- Best Pitch Award : INSPIRON 5.0, National Level Hackathon organized by CSI, COEP (2026)
- Meritorious Student Award : New Institute of Technology (2025)
- Ideathon Winner : Digifest 2024, New Institute of Technology